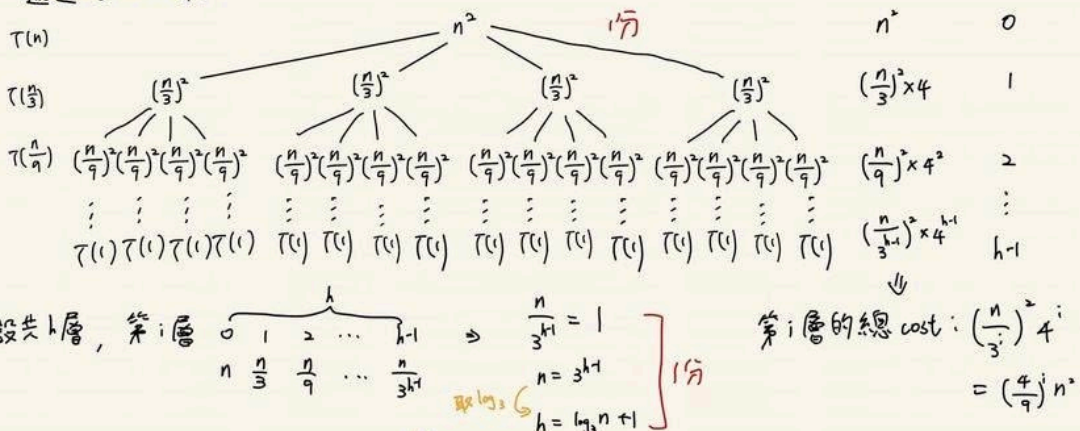


1. Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = 4T(\lfloor n/3 \rfloor) + n^2$, where $T(1) = 1$ and $T(2) = 2$. Use the substitution method to verify your answer.

遞迴樹找上界:



1分

$$T(n) = n^2 + \frac{4}{9}n^2 + (\frac{4}{9})^2 n^2 + \dots + (\frac{4}{9})^{\log_3 n} n^2 = \sum_{i=0}^{\log_3 n} (\frac{4}{9})^i n^2 < \sum_{i=0}^{\infty} (\frac{4}{9})^i n^2$$

(一定要有此式)

$$= \frac{1}{1 - \frac{4}{9}} n^2 = \frac{9}{5} n^2$$

$\Rightarrow T(n) = O(n^2)$

induction step:

(假設) 1分 設 $T(\lfloor \frac{n}{3} \rfloor) \leq c_2 (\frac{n}{3})^2$ for $\lfloor \frac{n}{3} \rfloor$

(代回) 1分

$$\begin{aligned} \Rightarrow T(n) &= 4T(\lfloor \frac{n}{3} \rfloor) + n^2 \\ &\leq 4c_2 (\frac{n}{3})^2 + n^2 \\ &\leq 4c_2 (\frac{n}{3})^2 + n^2 \\ &= \frac{4}{9} c_2 n^2 + n^2 \\ &\leq c_2 n^2 \end{aligned}$$

當 $c_2 = \frac{9}{5}$ 時, 恆成立

1分

substitution 驗證

Basic step:

1分

$$\begin{aligned} T(1) &= 1 \leq c_1 \times 1^2 \Rightarrow c_1 \geq 1 \\ T(2) &= 2 \leq c_1 \times 2^2 \Rightarrow c_1 \geq \frac{1}{2} \end{aligned}$$

1分 $\Rightarrow c_1 = 1, n_0 = 1$

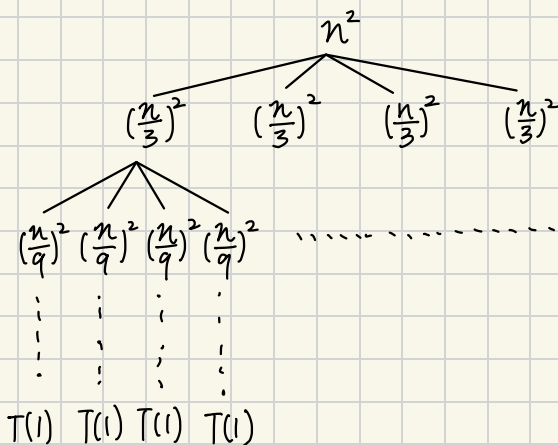
不必寫

$T(1)$ is true $\because$ from basic
 $T(2)$ is true $\because$ from basic
 $T(3)$ is true $\because$ from $T(1)$ is true + induction step
 $\vdots$
 $T(n)$ is true $\because$ from $T(\lfloor \frac{n}{3} \rfloor)$ is true + induction step

取 $n_0 = 1, c = \max \{c_1, c_2\} = \frac{9}{5}$

1. Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = 4T(\lfloor n/3 \rfloor) + n^2$, where $T(1) = 1$ and $T(2) = 2$. Use the substitution method to verify your answer.

$$T(n) = 4T(\lfloor \frac{n}{3} \rfloor) + n^2 \quad / \quad T(1)=1, T(2)=2$$



每層花費

$$n^2$$

層數

$$0$$

$$(\frac{n}{3})^2 \times 4$$

$$1$$

$$(\frac{n}{9})^2 \times 4^2$$

$$2$$

$$\vdots$$

$$\vdots$$

$$(\frac{n}{3^{h-1}})^2 \times 4^{h-1}$$

$$h-1$$

等比級數至 n 項之和:

$$S_n = \frac{a_1(r^n - 1)}{r - 1} \quad (r \neq 1)$$

* 無窮等比級數和:

When $|r| < 1$, 級數收斂
則存在和:

$$S = \frac{a}{1-r}$$

假設遞迴樹總共 h 層:

層數	0	1	2		$h-1$
子問題大小	n	$\frac{n}{3}$	$\frac{n}{9}$		$\frac{n}{3^{h-1}}$

$$\Rightarrow \frac{n}{3^{h-1}} = 1, n = 3^{h-1}, h = \log_3 n + 1$$

- 一直到 $h-1$

$$\text{第 } k \text{ 層 cost: } (\frac{n}{3^k})^2 \times 4^k = (\frac{4}{9})^k \times n^2$$

Total cost:

$$T(n) = n^2 + \frac{4}{9}n^2 + (\frac{4}{9})^2n^2 + \dots + (\frac{4}{9})^{\log_3 n}n^2$$

$$= \sum_{i=0}^{\log_3 n} (\frac{4}{9})^i n^2 < \sum_{i=0}^{\infty} (\frac{4}{9})^i n^2 = \frac{1}{1 - \frac{4}{9}} n^2 = \frac{9}{5}n^2$$

CH4 P18

$$\Rightarrow T(n) = O(n^2)$$

substitution method 驗證 (💡 O -notation: such that $f(n) \leq cg(n) \quad \forall n \geq n_0 \xrightarrow{\text{key pf}} T(n) \leq cn^2$)

Basic Step:

$$T(1) = 1 \leq C_1 \times 1^2 \Rightarrow C_1 \geq 1$$

$$T(2) = 2 \leq C_1 \times 2^2 \Rightarrow C_1 \geq \frac{1}{2}$$

$$\Rightarrow C_1 = 1, n_0 = 1$$

$$\therefore \exists C = \max(C_1, C_2) = \frac{9}{5} > 0, n_0 = 1$$

$$\text{s.t. } T(n) \leq Cn^2, \forall n \geq n_0$$

Induction Step

$$\text{assume } T(\lfloor \frac{n}{3} \rfloor) \leq C_2 (\frac{n}{3})^2 \text{ for } \lfloor \frac{n}{3} \rfloor$$

$$\text{then, } T(n) = 4T(\lfloor \frac{n}{3} \rfloor) + n^2$$

$$\leq 4C_2 (\frac{n}{3})^2 + n^2$$

$$\leq 4C_2 (\frac{n}{3})^2 + n^2$$

$$= \frac{4}{9}C_2 n^2 + n^2$$

$$\leq C_2 n^2$$

$$\frac{4}{9}C_2 n^2 + n^2 \leq C_2 n^2$$

$$\frac{4}{9}C_2 + 1 \leq C_2$$

$$1 \leq \frac{5}{9}C_2$$

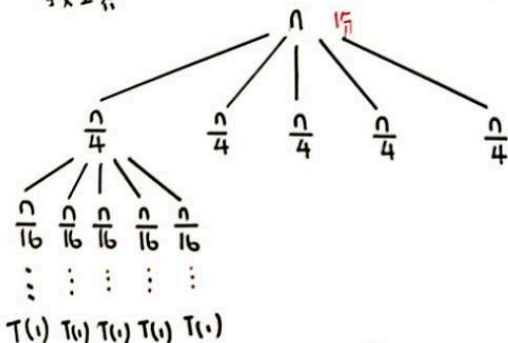
$$C_2 \geq \frac{9}{5}$$

當 $C_2 \geq \frac{9}{5}$ 時, 恆成立

姓名：
學號：A311
系所：

1. Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = 5T(\lfloor n/4 \rfloor) + n$, where $T(1) = 1$, $T(2) = 2$, and $T(3) = 3$. Use the substitution method to verify your answer.

遞迴樹
找上界：



每層總花費	層數
n	0
$\frac{n}{4} \cdot 5$	1
$\frac{n}{16} \cdot 25$	2
$\vdots$	$\vdots$
$\frac{n}{4^{h-1}} \cdot 5^{h-1}$	$h-1$

設總共 h 層，第 i 層 $\overbrace{0 \ 1 \ 2 \ \dots \ h-1}^{\pm h \text{ 層}}$
問題大小 $n \ \frac{n}{4} \ \frac{n}{16} \ \dots \ \frac{n}{4^{h-1}}$ 最後一層大小為 1 $\Rightarrow n = 4^{h-1} \Rightarrow h = \log_4 n + 1$

第 i 層總花費： $(\frac{n}{4^i}) \cdot 5^i = (\frac{5}{4})^i \cdot n$
 總花費 $T(n) = n + (\frac{5}{4})n + (\frac{5}{4})^2 n + \dots + \frac{5}{4} \log_4 n \cdot n = \sum_{i=0}^{\log_4 n} (\frac{5}{4})^i \cdot n = \frac{1 - (\frac{5}{4})^{\log_4 n + 1}}{1 - \frac{5}{4}} \cdot n$
 $= [-4 + 4 \cdot (\frac{5}{4})^{\log_4 n + 1}] \cdot n = -4n + 5 \cdot \frac{n^{\log_4 5}}{n^{\log_4 4}} \cdot n = 5n^{\log_4 5} - 4n = O(n^{\log_4 5})$

Substitution 驗證 pf $\exists c_0, n_0$, s.t. $T(n) \leq c \cdot n^{\log_4 5} - 5n \ \forall n, n \geq n_0$

Basic step:

$T(1) = 1 \leq c_1 \cdot 1^{\log_4 5} - 5 \Rightarrow c_1 \geq 6$

$T(2) = 2 \leq c_1 \cdot 2^{\log_4 5} - 10 \Rightarrow c_1 \geq 6$

$T(3) = 3 \leq c_1 \cdot 3^{\log_4 5} - 15 \Rightarrow c_1 \geq 6$

$\Rightarrow c_1 = 6, n_0 = 1$

Induction step:

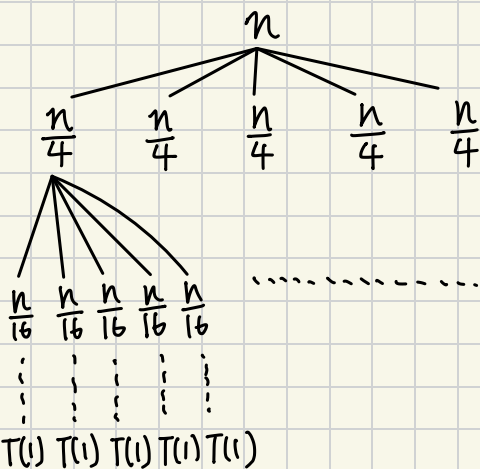
設 $T(\lfloor \frac{n}{4} \rfloor) \leq c_2 \cdot (\lfloor \frac{n}{4} \rfloor)^{\log_4 5} - 5 \lfloor \frac{n}{4} \rfloor$ for $\lfloor \frac{n}{4} \rfloor$

$\Rightarrow T(n) = 5T(\lfloor \frac{n}{4} \rfloor) + n$
 $\leq 5 \cdot [c_2 \cdot (\lfloor \frac{n}{4} \rfloor)^{\log_4 5} - 5 \lfloor \frac{n}{4} \rfloor] + n$
 $\leq 5 \cdot c_2 \cdot \frac{n^{\log_4 5}}{5^{\log_4 4}} - \frac{21}{4}n$
 $= c_2 \cdot n^{\log_4 5} - \frac{21}{4}n$
 $\leq c_2 \cdot n^{\log_4 5} - 5n \Rightarrow$ 當 $c_2 \geq 6$ 時恒成立

$\therefore \exists c = \max\{c_1, c_2\} = 6 \geq 6, n_0 = 1$
 s.t. $T(n) \leq 6 \cdot n^{\log_4 5} - 5n, \forall n, n \geq 1$

1. Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = 5T(\lfloor n/4 \rfloor) + n$, where $T(1) = 1$, $T(2) = 2$, and $T(3) = 3$. Use the substitution method to verify your answer.

$$T(n) = 5T(\lfloor \frac{n}{4} \rfloor) + n \quad / \quad T(1)=1, T(2)=2, T(3)=3$$



每層花費 層數
 n 0

$$(\frac{n}{4}) \times 5 \quad 1$$

$$(\frac{n}{16}) \times 5^2 \quad 2$$

$$(\frac{n}{4^{h-1}}) \times 5^{h-1} \quad h-1$$

第k層 cost:

$$(\frac{n}{4^k}) \times 5^k = (\frac{5}{4})^k \times n$$

Total cost:

$$n + \frac{5}{4}n + \frac{25}{16}n + \dots + (\frac{5}{4})^{\log_4 n} n$$

$$= \sum_{i=0}^{\log_4 n} (\frac{5}{4})^i n$$

$$= \frac{(\frac{5}{4})^{\log_4 n+1} - 1}{\frac{5}{4} - 1} \times n$$

$$= \frac{(\frac{5}{4})^{\log_4 n+1} - 1}{\frac{1}{4}} \times n = 4n[(\frac{5}{4})^{\log_4 n} \times (\frac{5}{4}) - 1]$$

$$= 4n \times (\frac{5^{\log_4 n}}{4^{\log_4 n}} \times \frac{5}{4} - 1) = 4n(\frac{n^{\log_4 5}}{n} \times \frac{5}{4} - 1)$$

$$= 4n(\frac{n^{\log_4 5}}{n} \times \frac{5}{4} - 1) = 4n(\frac{n^{\log_4 5}}{n} \times \frac{5}{4} - 1) = 5n^{\log_4 5} - 4n$$

$$\Rightarrow T(n) = O(n^{\log_4 5})$$

如果單純 $\leq Cn^{\log_4 5}$, 會有 +n 項消不掉

猜 -dn 項

substitution method 驗證

pf $\exists C, n_0 > 0$, s.t. $T(n) \leq Cn^{\log_4 5} - 5n \quad \forall n \geq n_0$

Basic Step:

$T(1) = 1 \leq C_1 \times 1^{\log_4 5} - 5 \Rightarrow C_1 \geq 6$

$T(2) = 2 \leq C_1 \times 2^{\log_4 5} - 10 \Rightarrow C_1 \geq 6$

$T(3) = 3 \leq C_1 \times 3^{\log_4 5} - 15 \Rightarrow C_1 \geq 6$

$\Rightarrow C_1 = 6, n_0 = 1$

$\therefore \exists C = \max(C_1, C_2) = 6, n_0 = 1$

s.t. $T(n) \leq Cn^2, \forall n \geq n_0$

Induction Step

assume $T(\lfloor \frac{n}{4} \rfloor) \leq C_2(\lfloor \frac{n}{4} \rfloor)^{\log_4 5} - 5(\lfloor \frac{n}{4} \rfloor)$ for $\lfloor \frac{n}{4} \rfloor$

then, $T(n) = 5(\lfloor \frac{n}{4} \rfloor) + n$

$$\leq 5 \times [C_2(\lfloor \frac{n}{4} \rfloor)^{\log_4 5} - 5(\lfloor \frac{n}{4} \rfloor)] + n$$

$$\leq C_2 n^{\log_4 5} - \frac{25n}{4} + \frac{4n}{4}$$

$$= C_2 n^{\log_4 5} - \frac{21n}{4}$$

$\leq C_2 n^{\log_4 5} - 5n \Rightarrow$ 當 $C_2 \geq 0$ 時, 恆成立

姓名:

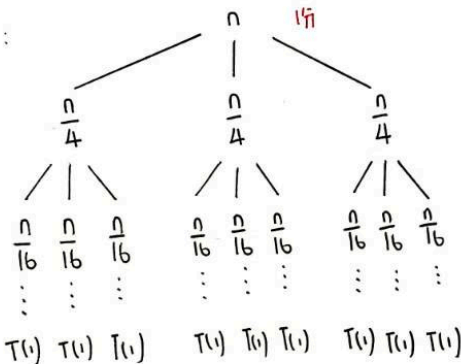
學號: B班

系所:

1. Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = 3T(\lfloor n/4 \rfloor) + n$, where $T(1) = 1$, $T(2) = 2$, and $T(3) = 3$. Use the substitution method to verify your answer.

3個子問題
每個大小是原本的 $\frac{1}{4}$

遞迴樹:
找上界



(共h層)
層數
每層總花費
n
0
 $\frac{n}{4} \cdot 3$ 1
 $\frac{n}{16} \cdot 9$ 2
 $\vdots$
 $\frac{n}{4^{h-1}} \cdot 3^{h-1}$ h-1

設總共h層, 第i層

子問題大小 $n \quad \frac{n}{4} \quad \frac{n}{16}$

$\frac{n}{4^{h-1}}$ 最後一層大小為1 $\Rightarrow n = 4^{h-1}$

第i層的總花費: $(\frac{n}{4}) \cdot 3^i = (\frac{3}{4})^i \cdot n$

取 $\log_4$ $h = \log_4 n + 1$

總 cost $T(n) = n + \frac{3}{4}n + (\frac{3}{4})^2 n + \dots + (\frac{3}{4})^{\log_4 n} n = \sum_{i=0}^{\log_4 n} (\frac{3}{4})^i \cdot n < \sum_{i=0}^{\infty} (\frac{3}{4})^i \cdot n = \frac{1}{1-\frac{3}{4}} \cdot n = 4n$

$\Rightarrow T(n) = O(n)$

Substitution 驗證:

Basic step:

$T(1) = 1 \leq C_1 \cdot 1 \Rightarrow C_1 \geq 1$
 $T(2) = 2 \leq C_1 \cdot 2 \Rightarrow C_1 \geq 1$
 $T(3) = 3 \leq C_1 \cdot 3 \Rightarrow C_1 \geq 1$

$\Rightarrow C_1 = 1, n_0 = 1$

Induction step: 假設 $T(\frac{n}{4})$ 成立
pf. $T(n)$ 成立

17 設 $T(\frac{n}{4}) \leq C_2 \cdot \frac{n}{4}$ for $\frac{n}{4}$

$\Rightarrow T(n) = 3T(\frac{n}{4}) + n$

$\leq 3 \cdot C_2 \cdot \frac{n}{4} + n$

$\leq 3 \cdot C_2 \cdot \frac{n}{4} + n$

$= \frac{3}{4}C_2 n + n \leq C_2 n$

當 $C_2 \geq 4$ 時, 恆成立

$n \leq \frac{1}{4}C_2 n$
 $\Rightarrow 1 \leq \frac{1}{4}C_2$
 $\Rightarrow C_2 \geq 4$

17 $\therefore \exists C = \max\{C_1, C_2\} = 4, n_0 = 1$ s.t. $T(n) \leq C \cdot n, \forall n, n \geq n_0$

1. Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = 3T(\lfloor n/4 \rfloor) + n$, where $T(1) = 1$, $T(2) = 2$, and $T(3) = 3$. Use the substitution method to verify your answer.

$$T(n) = 3T(\lfloor \frac{n}{4} \rfloor) + n \quad / \quad T(1) = 1, T(2) = 2, T(3) = 3$$

每層花費 層數
 n 0

$$(\frac{n}{4}) \times 3$$

$$(\frac{n}{16}) \times 3^2$$

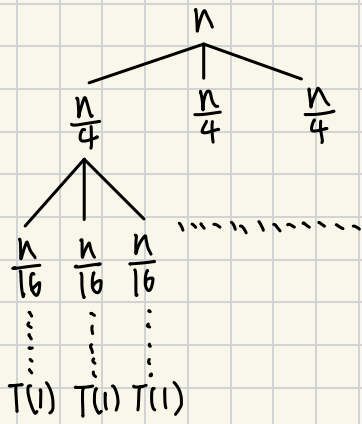
$$\frac{n}{4^{h-1}} \times 3^{h-1}$$

假設共 h 層

層數 0 1 2 ... $h-1$

問題大小 $n \quad \frac{n}{4} \quad \frac{n}{16} \quad \dots \quad \frac{n}{4^{h-1}}$

$$\Rightarrow \frac{n}{4^{h-1}} = 1, h = \log_4 n + 1$$



第 k 層 cost: $\frac{n}{4^k} \times 3^k = (\frac{3}{4})^k \times n$

Total cost: $T(n) = n + \frac{3}{4}n + (\frac{3}{4})^2n + \dots + (\frac{3}{4})^{\log_4 n}n = \sum_{i=0}^{\log_4 n} (\frac{3}{4})^i n$

$$\sum_{i=0}^{\log_4 n} (\frac{3}{4})^i n < \sum_{i=0}^{\infty} (\frac{3}{4})^i n = \frac{n}{1 - \frac{3}{4}} = 4n, T(n) \Rightarrow O(n)$$

substitution 驗證

Basic Step:

$$T(1) = 1 \leq C_1 \times 1, C_1 \geq 1$$

$$T(2) = 2 \leq C_1 \times 2, C_1 \geq 1$$

$$T(3) = 3 \leq C_1 \times 3, C_1 \geq 1$$

$$\Rightarrow C_1 = 1, n_0 = 1$$

Induction Step:

assume $T(\lfloor \frac{n}{4} \rfloor) \leq C_2 (\lfloor \frac{n}{4} \rfloor)$ for $\lfloor \frac{n}{4} \rfloor$

then, $T(n) = 3T(\lfloor \frac{n}{4} \rfloor) + n$

$$\leq 3C_2 (\lfloor \frac{n}{4} \rfloor) + n$$

$$\leq 3C_2 (\frac{n}{4}) + n$$

$$= \frac{3}{4}C_2 n + n$$

$$\leq C_2 n$$

當 $C_2 \geq 4$ 時, 恆成立

$$\frac{3}{4}C_2 n + n \leq C_2 n$$

$$\frac{3}{4}C_2 + 1 \leq C_2$$

$$1 \leq \frac{1}{4}C_2$$

$$C_2 \geq 4$$

$$\therefore \exists C = \max(C_1, C_2) = 4, n_0 = 1,$$

$$\text{s.t. } T(n) \leq Cn \quad \forall n \geq n_0$$

評分標準:

Recursion Tree Method: 4分

- Draw recursion Tree 1分
- 每層總花費表格 1分
- 算樹高 h 1分
- 算出 Upper Bound 1分

Substitution Method: 6分

- Basic Step 證明 1分
- 清楚寫出 C_1, n_0 1分
- Induction Step 假設、計算代回 2分
- C_2 結論 1分 (當 $C_2 \geq ?$ 時恆成立)
- 證明結論 1分 $\left(\begin{array}{l} \therefore \exists C = \max(C_1, C_2) = ? > 0, n_0 = ? \\ \text{s.t. } T(n) \leq \underline{\hspace{2cm}} \quad \forall n \geq n_0 \end{array} \right)$